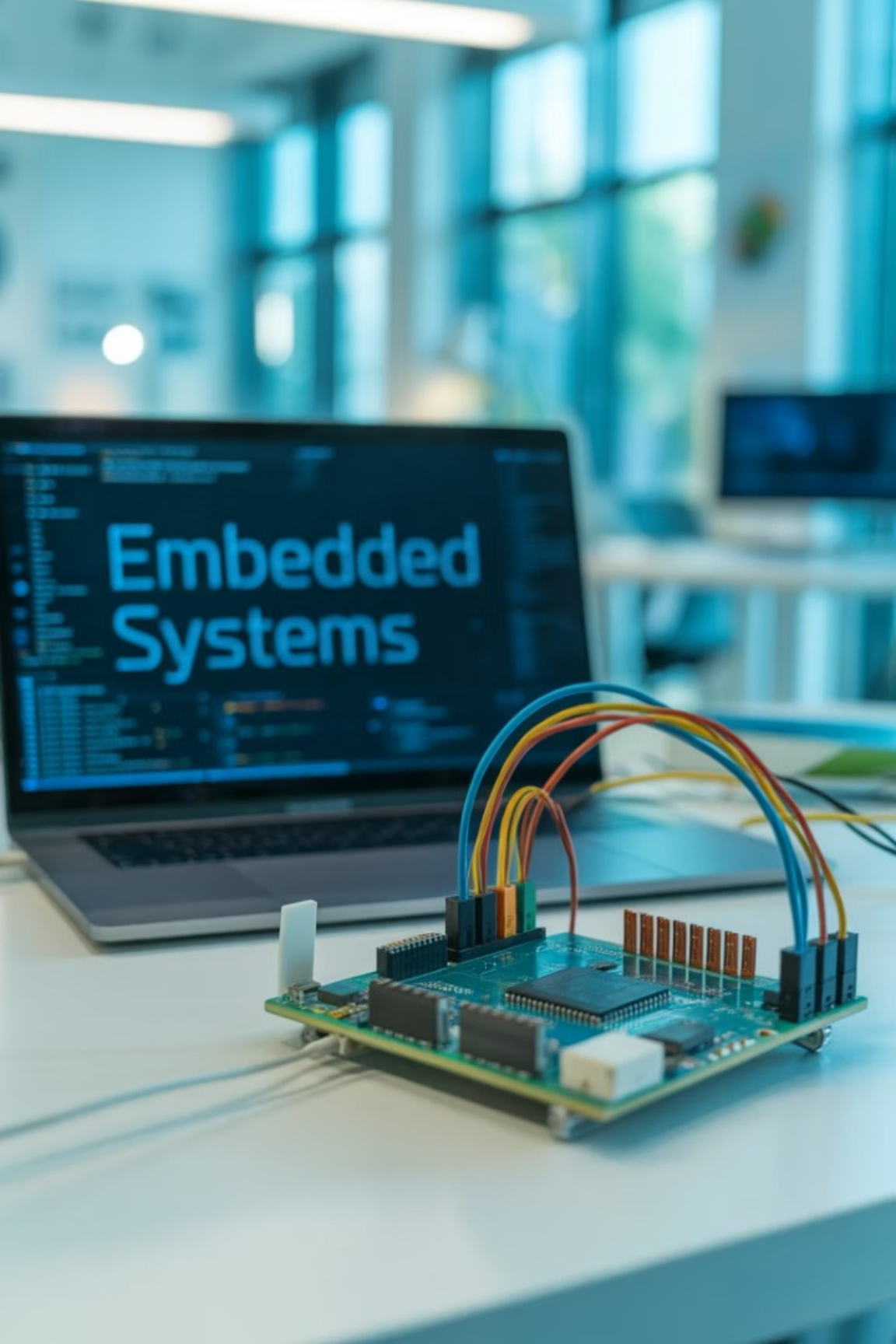




# Zephyr RTOS Intro





## Achieve Your Goals

# Agenda

1

## What is Zephyr RTOS?

Small, scalable open-source RTOS

2

## Key Features

Architecture, boards, connectivity

3

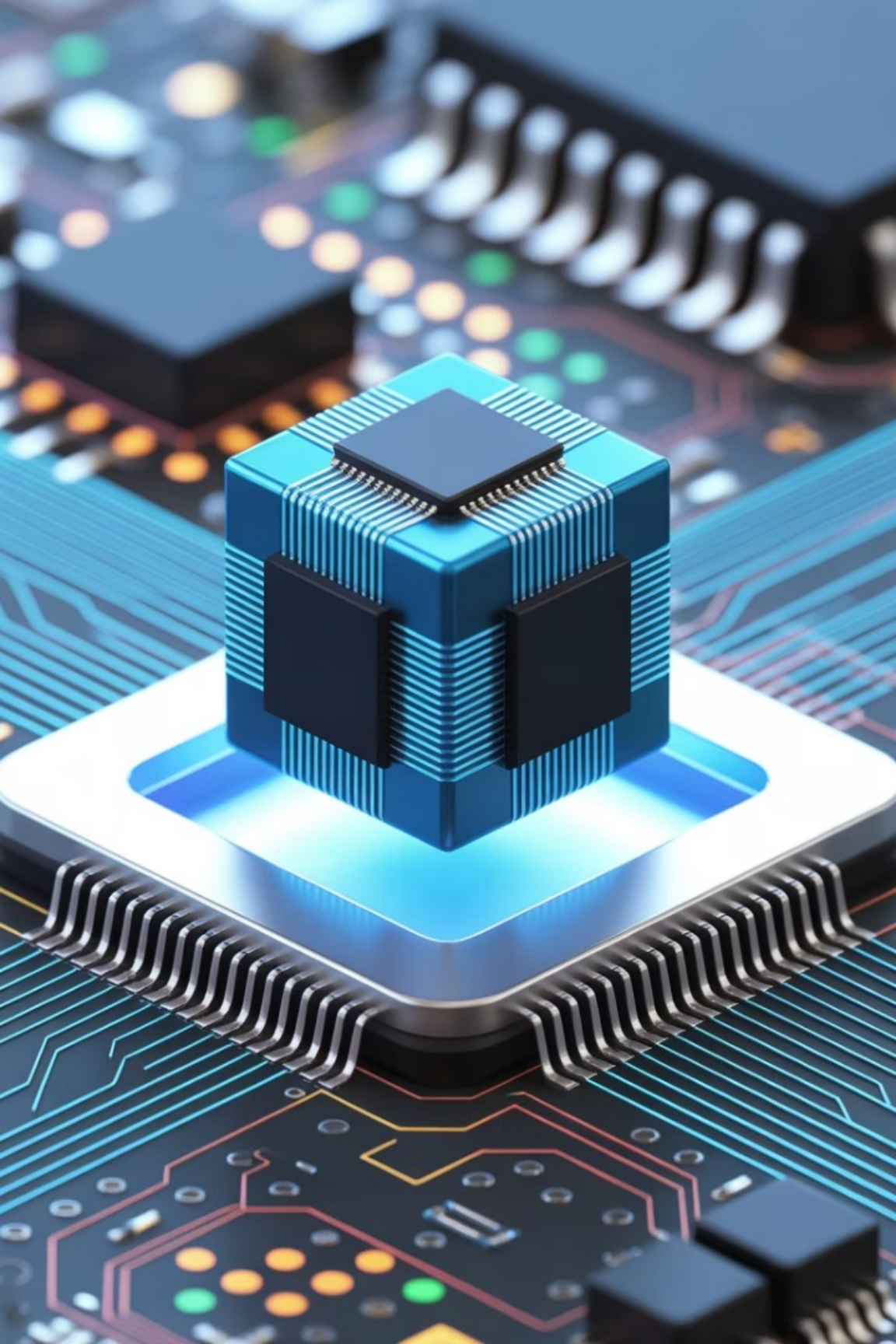
## Advantages

Compared to alternatives

4

## Real-World Use

Products, trends, getting started



# What is Zephyr RTOS?

## Small & Scalable

Designed for resource-constrained embedded systems

## Open Source

Vendor-neutral, community-driven

## Wide Support

ARM, RISC-V, x86, Xtensa

## Heritage

Originated from Wind River's Rocket OS (2015)

# Key Features

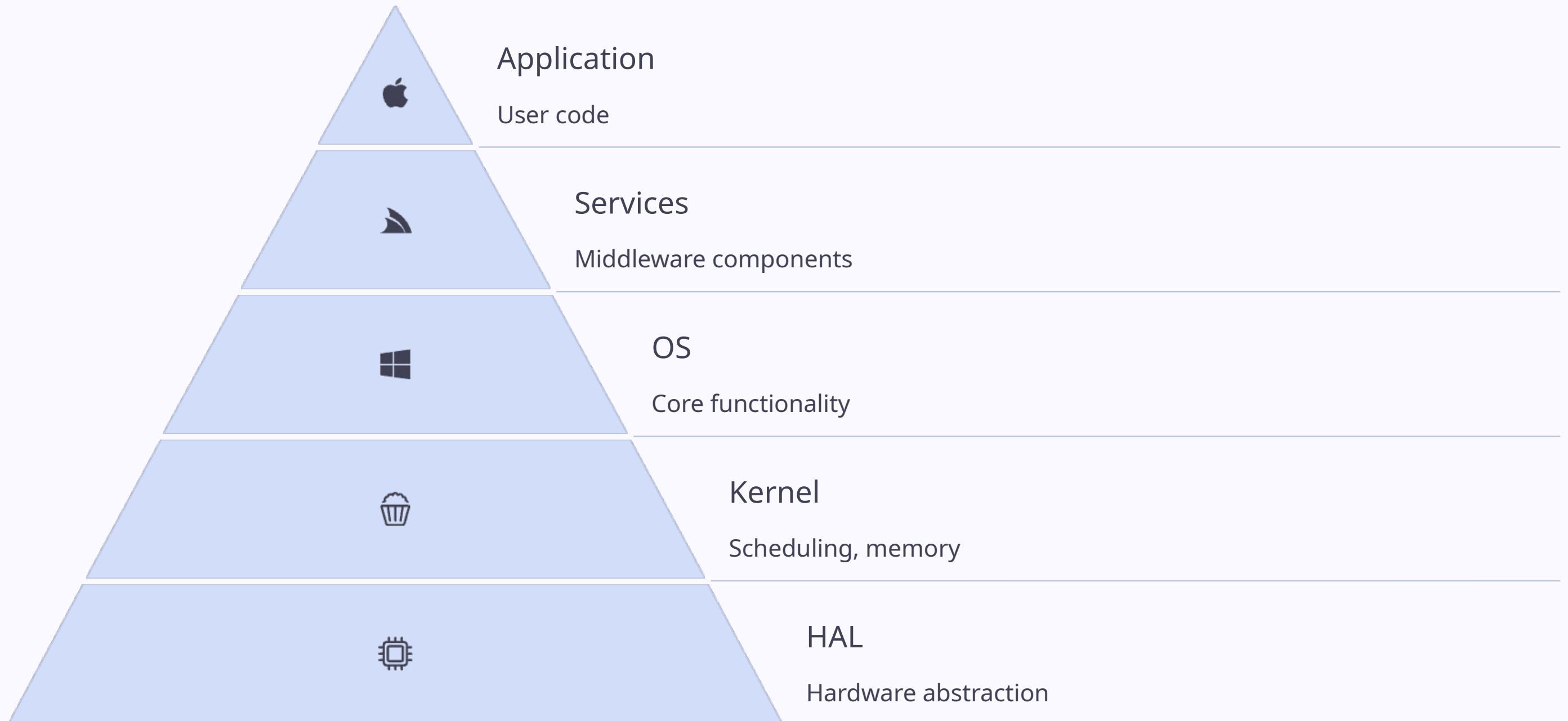
## Minimal Footprint

- <8KB Flash, <5KB RAM
- Scalable to multi-core
- BLE 5.3, Wi-Fi, CAN, USB

## Extensive Support

- 750+ boards, 220+ sensors
- CMake, Kconfig, Devicetree
- Full protocol support
- Power management

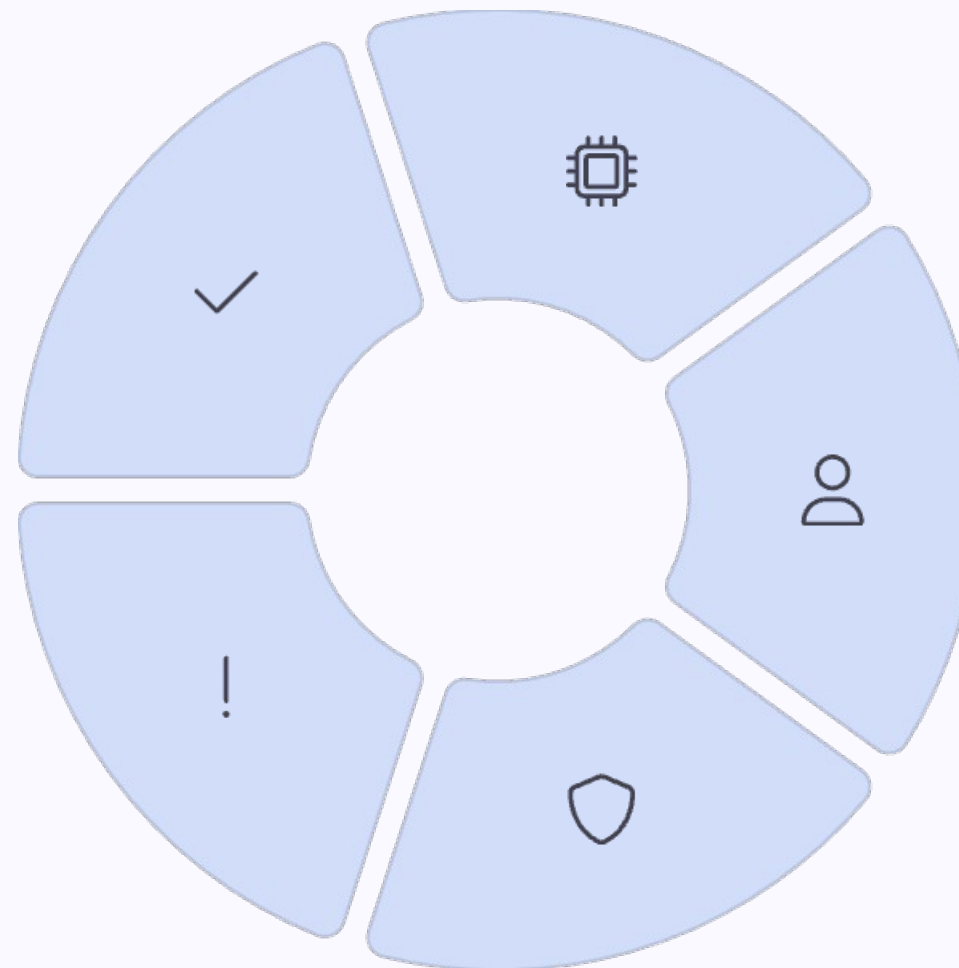
# Architecture Overview



# Zephyr vs Other RTOSes

Open Source  
Apache 2.0 vs commercial licensing

Challenges  
Steeper learning curve, newer ecosystem



## Hardware Support

750+ boards vs 40-100 for others

## Community

1,100+ contributors, 5x growth since 2017

## Security

Dedicated PSIRT and vulnerability management



# Advantages over Alternatives

| Feature     | Zephyr RTOS  | Linux   | Bare Metal |
|-------------|--------------|---------|------------|
| Memory      | KBs          | MBs     | Minimal    |
| Boot Time   | Milliseconds | Seconds | Immediate  |
| Determinism | High         | Low     | Manual     |
| Portability | High         | Medium  | None       |
| Power       | Efficient    | Hungry  | Variable   |

# Real-World Products Using Zephyr



## Consumer

Galaxy Ring, Moto Watch 100



## Computing

Chromebook EC, Framework Laptop



## Industrial

Vestas, BLiXT, IoT devices

25+ Million Devices in the Field



# Getting Started



## Setup

Install deps, setup west, get SDK



## Develop

Clone repo, choose sample, target board



## Deploy

Build/flash/debug with west



## Learn More

Docs, GitHub, Discord